

PHASE 1

Introduction and Literature Review

Chapters 1-2: background, problem, objectives, scope and existing systems

FITBUDDY • AI FITNESS PLAN GENERATOR • PROJECT REPORT

Chapter 1: Introduction

1.1 Background

Digital technology has changed how people approach physical fitness. Mobile applications, wearable devices, online video programmes and web platforms now make exercise instructions and wellness information available to a very large audience. This growth has lowered the barrier to entry for people who want to begin or maintain a fitness routine, but it has also created a new difficulty: the volume of available guidance is far greater than any individual can evaluate, and most of it is written for a general audience rather than for a specific person.

Effective exercise planning depends on individual factors. Age, body weight, current fitness level, the desired outcome (for example fat loss, muscle gain or general endurance) and the intensity a person is prepared to sustain all influence what a suitable weekly routine looks like. A programme that is appropriate for a young adult targeting muscle gain may be inappropriate for an older beginner who wants to improve general mobility. Generic plans published on websites or in printed material cannot account for these differences, and people who follow them often abandon the routine because it is too hard, too easy or simply irrelevant to their goals.

Recent progress in artificial intelligence, and in particular in large language models (LLMs), offers a practical way to close this gap. Generative models can accept a short natural-language description of a person, reason over the constraints it contains and produce a structured, readable plan in a few seconds. When such a model is embedded in a well-designed web application with validation, persistent storage and a feedback loop, the result is a tool that behaves less like a static document and more like an adaptive planning assistant. FitBuddy is an implementation of this idea for weekly workout planning and basic nutrition and recovery guidance.

1.2 Problem Statement

Creating a structured weekly workout plan normally requires either professional coaching or a good deal of personal knowledge and effort. Most users have neither. They therefore rely on generic programmes, which introduces several problems:

- Generic workout plans are written for an average user. They do not reflect the age, weight, goal or preferred intensity of the person following them.
- Static fitness websites present fixed content. The user must search, compare and adapt the material manually, and the content does not change after the user reports that it did not work.
- Manual planning demands knowledge of exercise selection, volume, rest and progression. Errors in manual planning can produce ineffective or excessively demanding routines.
- Lack of adaptive feedback means that a plan, once obtained, cannot be revised in response to comments such as "the workouts are too long" or "I do not have access to a gym".

The problem addressed by this project is therefore how to provide personalised, structured and revisable workout guidance, together with brief nutrition and recovery advice, without requiring the user to design the plan by hand, and to do so in a system that stores the results and allows them to be reviewed by an administrator or coach.

1.3 Motivation

FitBuddy was developed to demonstrate that a modern generative AI service can be integrated into a conventional web application stack in a disciplined way. Three motivations guided the work. First, personalised guidance is more likely to be followed than generic guidance, so there is a genuine user benefit in generating plans from profile data. Second, the feedback loop, in which a user comments on a plan and receives a revised version, is a natural fit for the interactive nature of LLMs but is rarely available in static systems. Third, the project offered an opportunity to apply several core engineering topics in a single system: REST-style API design, data validation, object-relational mapping, prompt design, structured output handling, responsive interface design and basic security practice.

1.4 Objectives

The objectives of the project are as follows:

1. To generate a personalised 7-day workout plan from the user's name, user ID, age, weight, fitness goal and preferred intensity.
2. To generate a short nutrition and recovery tip that complements the workout plan.
3. To adapt the content of the plan to the stated fitness goal and workout intensity.
4. To allow users to submit feedback on a generated workout plan.
5. To generate a revised workout plan by analysing the original plan together with the user's feedback.
6. To preserve the original plan unchanged in the database so that the original and revised versions can be compared.
7. To store users, plans, feedback and revisions persistently using SQLite and SQLAlchemy.
8. To provide an administrative dashboard through which authorised administrators can monitor users, plans, feedback, nutrition tips and revised plans.
9. To provide interactive API documentation through FastAPI's automatically generated `/docs` interface.

1.5 Scope

In scope. The system covers personalised workout planning over a seven-day period; nutrition and recovery suggestions of a general wellness nature; AI-assisted revision of a plan based on free-text feedback; storage of user profile data, generated plans, feedback and revised plans; and an administrative view of the stored data. The interface is a responsive web application and the service is exposed through a documented HTTP API.

Out of scope. FitBuddy is a general wellness and educational tool. It does not perform medical diagnosis, does not prescribe treatment, does not assess injuries or medical conditions, and does not replace a physician, physiotherapist, registered dietitian or qualified personal trainer. It does not verify exercise form, does not integrate with wearable devices and does not provide clinical-grade nutrition planning. Users with medical conditions, injuries or special circumstances should seek professional advice before following any generated plan.

1.6 Significance

From an academic viewpoint, the project provides a worked example of integrating a generative AI service into a layered web architecture. It illustrates how to keep API credentials on the server, how to validate input before it reaches the model, how to request structured output so that it can be rendered and stored consistently, and how to design a revision workflow that never overwrites source data. These are transferable skills relevant to many AI-assisted applications beyond fitness.

From a practical viewpoint, the system reduces the effort needed to obtain an initial structured plan and allows the plan to be refined through conversation-like feedback. It also gives coaches or administrators a consolidated view of what has been generated, which supports oversight of AI output. The project does not claim to replace professional coaching; its value lies in making a reasonable first draft of a plan quickly available and easily adjustable.

Chapter 2: Literature Review and Existing System

2.1 Existing Fitness Applications

Conventional fitness applications and websites generally fall into a few broad categories: exercise libraries with descriptions or videos, workout trackers that log completed sessions, programme catalogues offering fixed multi-week routines, and calorie or activity counters. Many combine several of these functions. Their common characteristic is that the core content is authored in advance by people or organisations and then presented to all users, with limited customisation such as selecting a level or a target from a menu.

These applications are valuable in practice: they provide reliable, professionally reviewed content, social features and progress records. The observations in this chapter are general characteristics of the category and are not claims about any particular commercial product.

2.2 Existing Approach

The prevailing approach to workout guidance can be summarised in four points.

- Predefined workout plans. A fixed set of programmes is offered, typically categorised by goal and experience level.
- Static recommendations. Nutrition and recovery advice is published as articles or fixed tips that are identical for every reader.
- Manual personalisation. If a plan does not fit, the user must modify it by hand, which requires exercise knowledge, or consult a trainer.
- Limited adaptation. Adjustment is usually restricted to choosing a different predefined programme rather than revising the current one in response to comments.

2.3 Limitations of Existing Systems

The approach described above has several limitations relevant to this project:

- Lack of personalisation. Selecting from a small number of categories cannot capture the combination of age, weight, goal and intensity that defines an individual.
- Limited feedback adaptation. Systems that record user ratings seldom use free-text comments to produce a modified plan.
- Generic recommendations. The same advice is shown to users with different needs.
- Fragmented workout and nutrition guidance. Exercise planning and nutrition or recovery advice are frequently separate features that are not presented together as one coherent recommendation.
- Limited AI interaction. Where automation exists, it often consists of rule-based selection rather than open-ended natural-language generation and revision.

2.4 AI in Fitness Applications

Generative AI models, trained on large text corpora using the transformer architecture [8], are able to follow written instructions and produce fluent, structured text [9]. The Gemini family of models is one such family of multimodal models offered through a public API [10]. In a fitness setting, these capabilities are relevant in four ways.

- Natural-language interaction. Users can describe goals and concerns in ordinary language, and the system can respond in a readable form.
- Personalised recommendations. Profile attributes can be embedded in a prompt so that the generated plan reflects them.
- Dynamic plan generation. A new plan can be produced on demand rather than selected from a fixed catalogue.
- Feedback analysis. The model can read a user's comments alongside the existing plan and produce a modified plan that addresses them.

These capabilities also carry limitations. Generated text can be inaccurate or inconsistent, the model has no knowledge of a user's medical history, and the same prompt may yield different outputs on different occasions. Any AI-based fitness system must therefore include validation, safety wording and human oversight. These issues are discussed in Chapters 6 and 11. General public-health guidance on physical activity, such as that published by the World Health Organization [11], provides the kind of baseline against which generated advice can be reviewed.

2.5 Proposed System

FitBuddy is proposed as a web-based system that combines profile-driven plan generation, a separate nutrition and recovery tip, a feedback-driven revision loop, persistent storage and an administrative view. A Gemini model generates the seven-day workout plan, a faster Gemini Flash model generates the nutrition and recovery tip, and a further model call revises the plan when feedback is submitted. The original plan is never overwritten; revisions are stored as separate records. The backend uses FastAPI and SQLAlchemy with SQLite, which keeps the architecture simple and appropriate for an academic project.

2.6 Comparison

Table 2.1 compares the general characteristics of conventional systems with those of FitBuddy. The entries for conventional systems describe common patterns and should not be read as statements about specific products.

Table 2.1: Conventional Systems vs FitBuddy

Feature	Conventional System	FitBuddy
Personalised plan	Limited	Yes
AI generation	Usually limited	Yes
7-day planning	Yes / limited	Yes
Nutrition guidance	Limited	Yes
Feedback revision	Limited	Yes
Persistent storage	Varies	Yes
Admin dashboard	Varies	Yes